RL2 - Exercises

Exercise RL2.1. Building and understanding NFAs Solve exercise HMU 2.3.4 (copied below for your convenience).

! **Exercise 2.3.4:** Give nondeterministic finite automata to accept the following languages. Try to take advantage of nondeterminism as much as possible.

* a) The set of strings over alphabet $\{0, 1, \ldots, 9\}$ such that the final digit has appeared before.

b) The set of strings over alphabet $\{0, 1, \ldots, 9\}$ such that the final digit has *not* appeared before.

c) The set of strings of 0's and 1's such that there are two 0's separated by a number of positions that is a multiple of 4. Note that 0 is an allowable multiple of 4.

Exercise RL2.2. NFAs to DFAs Solve exercise HMU 2.3.3 (copied below for your convenience).

! Exercise 2.3.3: Convert the following NFA to a DFA and informally describe the language it accepts.

	0	1
$\rightarrow p$	$\{p, q\}$	$\{p\}$
q	$\{r, s\}$	$\{t\}$
r	$\{p, r\}$	$\{t\}$
$*s$	$\emptyset$	$\emptyset$
$*t$	$\emptyset$	$\emptyset$

Exercise RL2.3. On ϵ-NFAs Solve exercises HMU 2.5.1. In 2.5.1(b) consider strings of length 2 or less (instead 3 or less as the book says).

*** Exercise 2.5.1:** Consider the following ϵ-NFA.

	ϵ	a	b	c
$\rightarrow p$	$\emptyset$	$\{p\}$	$\{q\}$	$\{r\}$
q	$\{p\}$	$\{q\}$	$\{r\}$	$\emptyset$
$*r$	$\{q\}$	$\{r\}$	$\emptyset$	$\{p\}$

a) Compute the ϵ-closure of each state.

b) Give all the strings of length three or less accepted by the automaton.

c) Convert the automaton to a DFA.

Exercise RL2.4. On ϵ-NFAs Solve exercises HMU 2.5.3(b).

Exercise 2.5.3: Design ϵ-NFA's for the following languages. Try to use ϵ-transitions to simplify your design.

a) The set of strings consisting of zero or more a's followed by zero or more b's, followed by zero or more c's.

! b) The set of strings that consist of either 01 repeated one or more times or 010 repeated one or more times.

! c) The set of strings of 0's and 1's such that at least one of the last ten positions is a 1.

Exercise RL2.5. Correctness of the subset construction In class we have seen the proof of correctness of the subset construction, i.e. that the language of an NFA $N = (Q, \Sigma, \delta_N, q_0, F)$ is the same as the language of the DFA $D = (Q_D, \Sigma, \delta_D, \{q_0\}, F_D)$ obtained by the subset construction. The key property prove is $\forall q \in Q, w \in \Sigma^*.\delta_N^*(q, w) = \delta_D^*(\{q\}, w)$. Prove this again on your own. HINT: Use induction on the length of w or alternatively on the structure of w.

Hungry of more exercises? Check the HMU book (exercises 2.2.X). Recall that Selected official solutions to exercises from HMU can also be found here: `http://infolab.stanford.edu/~ullman/ialcsols/sols.html`.